

MATH 215 – Fall 2004
First Exam Solutions

Problem 1. ($3 \times 5 = 15$ points) Consider the triangle in space with vertices $A(2, 0, 0)$, $B(2, 1, 0)$, and $C(0, 0, 1)$.

(a) Compute the angle at the vertex A , that is, the angle between the vectors $\vec{AB}$ and $\vec{AC}$.

Solution. $\vec{AB} = \langle 0, 1, 0 \rangle$ and $\vec{AC} = \langle -2, 0, 1 \rangle$, so

$$\vec{AB} \cdot \vec{AC} = 0.$$

Therefore $\vec{AB}$ and $\vec{AC}$ are perpendicular. □

Answer: (a) $\pi/2$ radians.

(b) Find the equation of the plane containing the triangle.

Solution. The cross product $\vec{AB} \times \vec{AC}$ is a normal vector. Since

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{vmatrix} = \langle 1, 0, 2 \rangle,$$

the equation of the plane is of the form $x + 2z = d$, where d is some constant. To find d , we use the fact that A is on the plane to find that $2 + 2(0) = d$, so $d = 2$. □

Answer: (b) $x + 2z = 2$

(c) Find the area of the triangle.

Solution. The area of the triangle is half the area of the parallelogram spanned by $\vec{AB}$ and $\vec{AC}$, which is

$$\frac{1}{2} |\vec{AB} \times \vec{AC}| = \frac{1}{2} \sqrt{1^2 + 0^2 + 2^2} = \frac{\sqrt{5}}{2}.$$

Answer: (c) $\frac{\sqrt{5}}{2}$

Problem 2. (5 + 10 = 15 points)

(a) Find parametric equations of the line through the points $A(-1, 2, 3)$ and $B(0, -1, 4)$, and give the values of the parameter corresponding to A and B .

Solution. There are several different valid ways to proceed. Here is one. Compute $\vec{AB} = \langle 1, -3, 1 \rangle$, and note that $\vec{OA} = \langle -1, 2, 3 \rangle$. Therefore a parametrization of the line is

$$\vec{r}(t) = t \langle 1, -3, 1 \rangle + \langle -1, 2, 3 \rangle.$$

□

Answer:

(a) $\langle x(t), y(t), z(t) \rangle = \langle t - 1, -3t + 2, t + 3 \rangle$. A corresponds to $t = 0$, B corresponds to $t = 1$.

(b) Let L and M be the lines with parametric equations $\vec{u}(t) = \langle -t + 1, 2t + 2, t + 3 \rangle$ and $\vec{v}(t) = \langle 2t, t + 4, -t + 4 \rangle$. Do these lines intersect? If they do, give the coordinates of the point of intersection.

Solution. We must set up a vector equation using *two* different parameters, one for each line. We should investigate if the following vector equation has solutions:

$$\vec{u}(t) = \vec{v}(s).$$

The unknowns are t and s . This vector equation is equivalent to a *system* of three scalar equations, namely:

$$\begin{cases} -t + 1 &= 2s \\ 2t + 2 &= s + 4 \\ t + 3 &= -s + 4 \end{cases}$$

Adding the first and the third equations implies $4 = s + 4$, so $s = 0$. Using the first equation again we obtain $t = 1$. *Now we must verify that $(s, t) = (0, 1)$ satisfies the second equation.* And it does! Therefore the lines do intersect. The point of intersection is given by

$$\vec{u}(1) = \langle 0, 4, 4 \rangle = \vec{v}(0).$$

□

Answer: (b) The lines do intersect, at $(0, 4, 4)$.

Problem 3. ($5 \times 4 = 20$ points) A golf ball is hit at time $t = 0$. Its position vector as a function of time is given by

$$\vec{r}(t) = \langle 2t, 3t, -t^2 + 4t \rangle.$$

Notice that at $t = 0$ the ball is at the origin of the coordinate system. The xy plane represents the ground. At some time $t_1 > 0$ the ball will return to the xy plane at some point $P(a, b, 0)$.

(a) Compute the velocity and the speed of the ball at an arbitrary time t .

Solution. Note the meaning of the mathematical language here. You are being asked to find the velocity and speed at *every* (arbitrarily given) time t , *not* at a time t which you arbitrarily specify.

The velocity is $\frac{d\vec{r}}{dt}$:

$$\vec{v}(t) = \frac{d\vec{r}}{dt}(t) = \langle 2, 3, -2t + 4 \rangle.$$

The speed at time t is the norm of this vector:

$$v(t) = |\vec{v}(t)| = \sqrt{2^2 + 3^2 + (-2t + 4)^2}.$$

Note that $s(t)$ is bad notation for the speed, since it already refers to arclength. □

Answer: (a) $\langle 2, 3, -2t + 4 \rangle, \sqrt{29 + 4t^2 - 16t}$

(b) Find the time $t_1 > 0$ and the coordinates of the point P where the ball hits the xy plane again.

Solution. t_1 is a solution to the equation $z(t) = 0$, i.e., the equation $-t^2 + 4t = 0$. The solutions to this equation are $t = 0$ and $t = 4$. Since $t_1 > 0$, we have $t_1 = 4$. □

Answer: (b) $t_1 = 4, P = (8, 12, 0)$.

(c) Set up a definite integral equal to the length of the arc of the trajectory from the origin to the point P . You do not have to evaluate the integral.

Solution. The distance is the integral of the speed from $t = 0$ to $t = t_1$: □

Answer: (c) $\int_0^4 \sqrt{2^2 + 3^2 + (-2t + 4)^2} dt$

(d) Find the equation of the vertical plane containing the trajectory.

Solution. There are (at least) two valid ways to approach this problem. The most elegant one is to start with the realization that a “vertical” plane has an equation of the form: $ax + by = c$. (This is because the normal vector to a vertical plane has z -component equal to zero.) But we know two points on the plane: at $t = 0$ the ball is at the origin, and at $t = t_1 = 4$ the ball is at $(8, 12, 0)$. Substituting these points into the equation we get the system:

$$\begin{cases} 0a + 0b = c \\ 8a + 12b = c \end{cases}$$

The first equation implies that $c = 0$; substituting this into the second gives $2a + 3b = 0$. This equation has infinitely many solutions, one being $a = 3$ and $b = -2$.

A more straightforward approach is to take three points on the trajectory and follow the usual approach to finding the equation of the plane through three non-collinear points. For example, you can take the points $O = (0, 0, 0)$, $Q = (2, 3, 3)$ and $P = (8, 12, 0)$ corresponding to $t = 0, 1, 4$

respectively. Notice that it is not *always* the case that the position vector $\vec{r}(t)$ of a point in the plane is parallel to the plane (although it is true in this context).

Then the vectors $\vec{OQ} = \langle 2, 3, 3 \rangle$ and $\vec{OP} = \langle 8, 12, 0 \rangle$ are parallel to the plane, so $\langle 2, 3, 3 \rangle \times \langle 8, 12, 0 \rangle = \langle -36, 24, 0 \rangle$ is a normal vector to the plane. Thus the equation of the plane is $-36x + 24y = d$ for some d . Using the fact that $(0, 0, 0)$ is one the plane, we get $-36(0) + 24(0) = d$, so $d = 0$ and an equation of the plane is $-36x + 24y = 0$. Simplifying gives $3x - 2y = 0$ as before.

Finally, note that we have found the vertical plane which *would* contain the trajectory, *if* the trajectory lay in such a plane. To verify that it does, we check that the coordinates of every point on the trajectory satisfy the equation of the plane:

$$3(2t) - 2(3t) + 0(-t^2 + 4t) = 0,$$

as desired. □

Answer: (d) $3x - 2y = 0$

Problem 4. (7 + 8 = 15 points) The temperature T of a mole of oxygen is given (in suitable units) in terms of its pressure P and volume V by the equation

$$T = \frac{16}{V} - \frac{1}{V^2} - P + 11VP.$$

(a) Find the temperature T and the differential dT if $V = 1$ and $P = 10$.

Solution. $dT = \frac{\partial T}{\partial V}dV + \frac{\partial T}{\partial P}dP$. Now $\frac{\partial T}{\partial V} = -\frac{16}{V^2} + \frac{2}{V^3} + 11P$, while $\frac{\partial T}{\partial P} = -1 + 11V$. Evaluating T , $\frac{\partial T}{\partial V}$ and $\frac{\partial T}{\partial P}$ at $(V, P) = (1, 10)$ gives: $\square$

Answer: (a) $T = 115$, $dT = 96dV + 10dP$

(b) Suppose the pressure increases to $P = 10.1$. Use your answer to part (a) to estimate how much the volume would have to change for the temperature to remain constant. Does the volume have to increase or decrease?

Solution. Remember that we have the differential approximation

$$\Delta T \approx \frac{\partial T}{\partial V}\Delta V + \frac{\partial T}{\partial P}\Delta P.$$

(Here ΔT , ΔV and ΔP measure actual physical changes in T , V and P , respectively, whereas dT , dV and dP above are just the names of variables.) We are given that $\Delta P = 10.1 - 10 = 0.1$, and require $\Delta T = 0$. Using the computations of part (a), we have

$$0 \approx 96\Delta V + 10 \cdot 0.1,$$

or $\Delta V \approx -\frac{1}{96}$. Since this is negative, the volume must decrease. $\square$

Answer: (b) The volume must decrease by approximately 1/96 units.

Problem 5. (5 + 5 + 10 = 20 points) Throughout this problem

$$f(x, y, z) = x^2 + 2y^2 + z^2/2 + 2xy.$$

The level surfaces of this function are ellipsoids.

(a) Find the directional derivative of f at $(-1, 1, 2)$ in the direction of the vector $3\vec{i} + 4\vec{k}$.

Solution. We will use the formula $D_{\vec{u}}f = \vec{u} \cdot \nabla f$. The gradient of f at (x, y, z) is

$$\nabla f(x, y, z) = \langle 2x + 2y, 4y + 2x, z \rangle.$$

Therefore $\nabla f(-1, 1, 2) = \langle 0, 2, 2 \rangle$. Now $\vec{u}$ should be the *unit* vector in the direction of $\langle 3, 0, 4 \rangle$, so, since $|\langle 3, 0, 4 \rangle| = \sqrt{25} = 5$, we should take

$$\vec{u} = \langle 3/5, 0, 4/5 \rangle.$$

Therefore, the directional derivative we want is

$$\langle 0, 2, 2 \rangle \cdot \langle 3/5, 0, 4/5 \rangle = 8/5.$$

□

Answer: (a) The directional derivative equals $8/5$.

(b) Find the direction (unit vector) in which the rate of change of f is greatest at the point $(-1, 1, 2)$, and the rate of change (scalar) in this direction.

Solution. The direction of greatest rate of change is the direction of the gradient, and the rate in that direction is the magnitude of the gradient. The direction therefore is given by the normalized gradient vector. One computes:

$$|\langle 0, 2, 2 \rangle| = \sqrt{8} = 2\sqrt{2}.$$

Since $\frac{2}{2\sqrt{2}} = \frac{\sqrt{2}}{2}$, the answers are:

□

Answer: (b) Direction: $\langle 0, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \rangle$. Rate: $2\sqrt{2}$.

(c) Find all the points on the level surface $f(x, y, z) = 4$ where the tangent plane to the surface is parallel to the xz plane.

Solution. The tangent plane to the level surface at any point (x, y, z) is perpendicular to $\nabla f(x, y, z)$. Therefore, the tangent plane is parallel to the xz plane at all points where $\nabla f(x, y, z)$ is perpendicular to the xz plane, that is, *at all points where $\nabla f(x, y, z)$ is parallel to $\vec{j}$* . (Note that $\nabla f(x, y, z)$ need not be *equal* to $\vec{j}$.) This condition translates into the system:

$$\begin{cases} 2x + 2y &= 0 \\ z &= 0 \end{cases}$$

obtained by setting the x - and y -components of ∇f equal to zero. Substituting $y = -x$ and $z = 0$ into the equation $f(x, y, z) = 4$, we obtain

$$x^2 + 2x^2 + 0 - 2x^2 = 4, \quad \text{which holds iff } x = \pm 2.$$

□

Answer: (c) $(2, -2, 0), (-2, 2, 0)$.

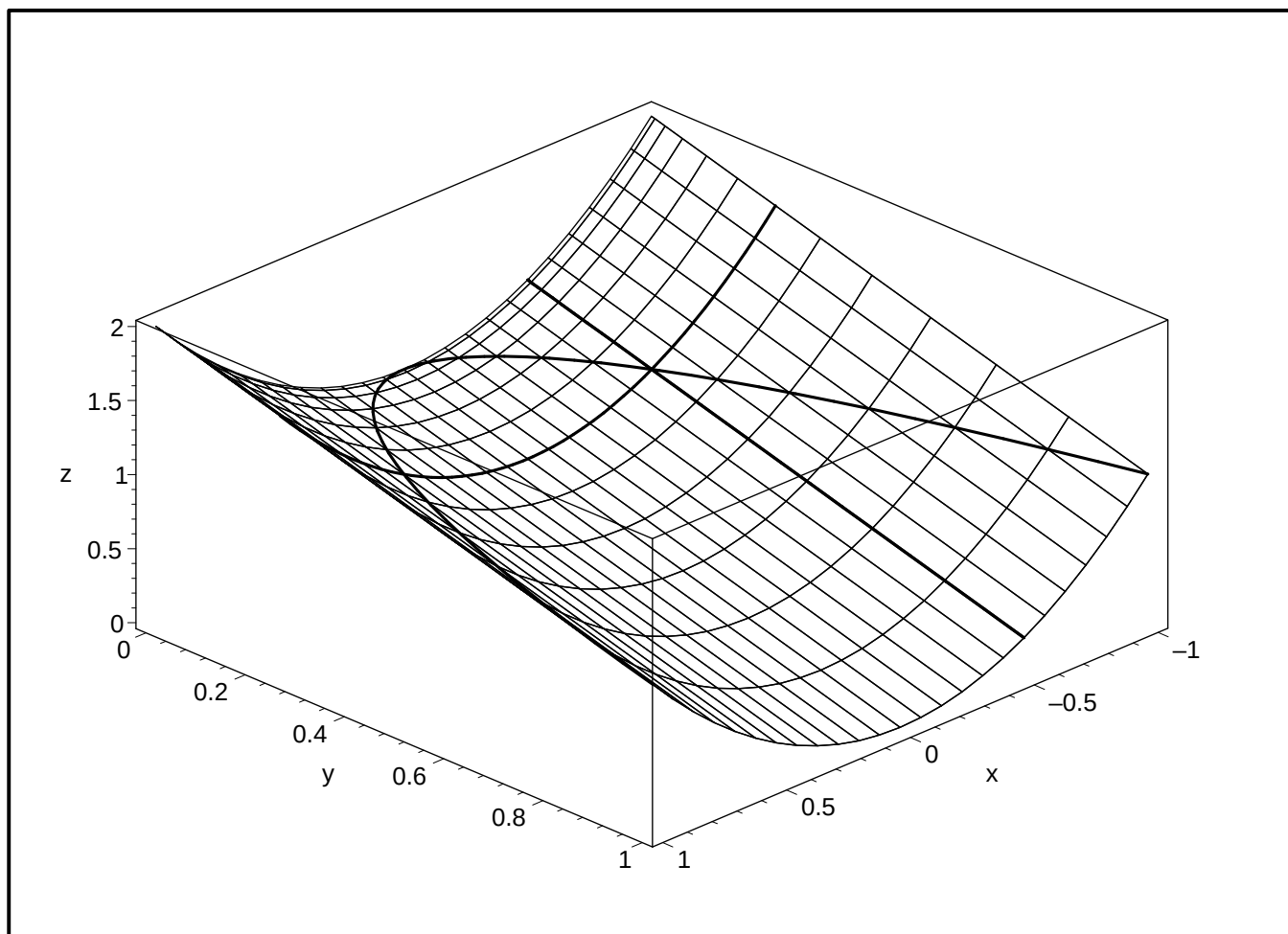
Problem 6. (15 points) (a) The graph below is a graph of the parametric surface

$$S_1 : \vec{r}(s, t) = \langle s, t^2, 1 - t^2 + s^2 \rangle, \quad -1 \leq t \leq 1, \quad -1 \leq s \leq 1.$$

The following curves have been drawn on the surface:

- (1) The curve C_1 corresponding to $s = -1/2$, $-1 \leq t \leq 1$.
- (2) The curve C_2 corresponding to $t = 1/2$, $-1 \leq s \leq 1$.
- (3) The curve C_3 corresponding to $s = t$, $-1 \leq t \leq 1$.

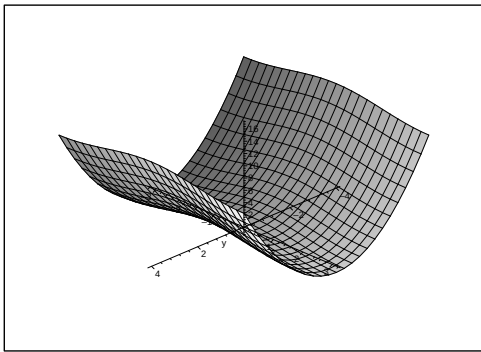
On the graph below **clearly label the corresponding curves** C_1 , C_2 , C_3 . *Ambiguous labelings will receive no credit.*



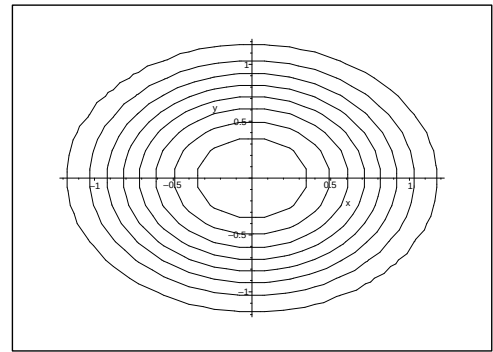
Solution. C_3 is the parabola. Along C_1 x is constant; along C_2 y is constant. □

(b) Match the graphs on the next page with the corresponding plots of level curves:

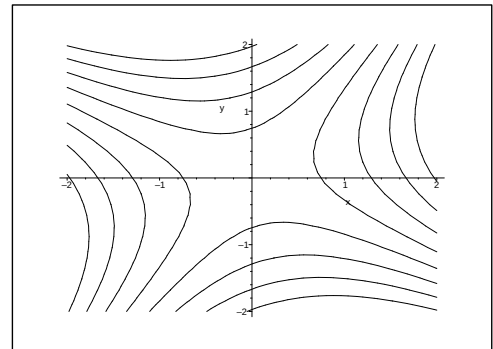
Graph No.	Level curves
1	C
2	A
3	B



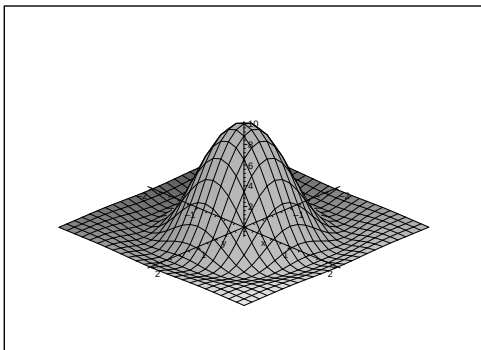
Graph 1



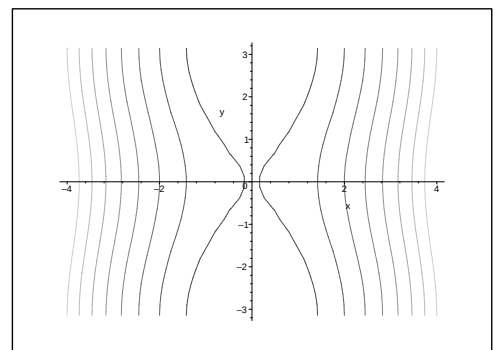
Levels A



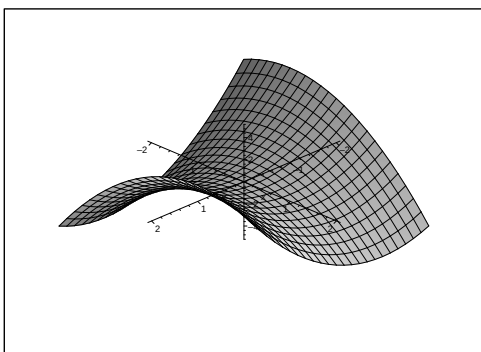
Levels B



Graph 2



Levels C



Graph 3